

SEBASTIAN R. KELTZ

Orlando, FL | 954-213-7892 | sebastiankeltz@icloud.com | [LinkedIn](#) | [Portfolio](#)

EDUCATION

University of Central Florida Orlando, FL	June 2027-Present
Bachelor of Science in Electrical Engineering	GPA: 3.11
Relevant Coursework: Circuit Analysis, Electronics, Electronics Laboratory, Power Systems Analysis, Linear Control Systems, Fundamentals of Electric Power	
Florida Gulf Coast University Fort Myers, FL	December 2024
Associate of Science in Software Engineering	GPA: 3.34

TECHNICAL SKILLS

- Programming:** C, MATLAB, GNU Octave
- Engineering Software:** PowerWorld Simulator, LTspice, Autodesk Fusion, Arduino IDE
- Computer Architecture Tools:** JArmEmu, MARS MIPS Simulator
- Hardware & Laboratory:** ESP32-S3, I²C sensors, breadboards, oscilloscopes, digital multimeters, function generators
- Technical Areas:** Circuit analysis, power-system simulation, embedded systems, sensor integration, power electronics, hardware testing

RELEVANT EXPERIENCE

Meat Cutter

Sprouts Farmers Market- Estero, Miami, Orlando, FL	May 2023-December 2025
Provided high-quality customer service while assisting guests with product selection and special requests	

Front Desk Service Agent

Naples Grande Beach Resort- Naples, FL	June 2022-December 2022
Communicated clearly with guests to resolve issues and enhance overall experience	

Eagle Scout

Boy Scouts of America- Miami, FL	August 2007-December 2021
- Led and managed a community service project benefiting local community - Demonstrated leadership, teamwork, and long-term commitment through achievement of Eagle Scout rank	

ENGINEERING PROJECTS

ESP32 Smart-Home Monitoring System (Independent Project)	2026–Present
- Developed modular C++ firmware for an ESP32-S3 and integrated a BME280 environmental sensor over I²C, verifying live temperature, humidity, and pressure measurements through serial diagnostics. - Simulated a MOSFET-controlled 12 V fan circuit in LTspice and analyzed flyback-diode protection, reducing the simulated drain-voltage transient from approximately 551 V to 12.4 V.	
Electric Three-Wheel Cooler Scooter (Independent Project)	September 2025-January 2026
- Designed and built a three-wheel electric scooter using two 18 V, 8 Ah batteries in series to create a 36 V power system, evaluating voltage, current, torque, and runtime. - Integrated a DC motor, motor controller, and throttle; measured voltage and current under load to verify stable power delivery and operation.	
Electric Dirt Bike Conversion (Independent Project)	October 2025-January 2026
- Converted a dirt bike to electric drive using a 48 V, 20 Ah lithium battery pack and 2,000 W BLDC motor, designing the power system for an estimated 40–45 A full-load current. - Integrated the motor controller, throttle, high-current wiring, protection components, and drivetrain hardware; tested the completed system under load and refined throttle response and electrical reliability.	